

PRAVEEN MAMIDIPALLI

AI/ML Developer — Python Developer — LLM & RAG Engineer

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GitHub

Vijayawada, AP

PROFESSIONAL SUMMARY

AI/ML developer with hands-on experience building local LLM systems, Retrieval-Augmented Generation (RAG) pipelines, and conversational AI applications. Skilled in Python, Flask, OpenCV, Ollama, and modern AI frameworks. Developed production-style projects integrating computer vision, voice interaction, vector search, and large language models. Passionate about Machine Learning, Natural Language Processing, and Generative AI.

TECHNICAL SKILLS

Programming Languages: Python, SQL

AI/ML: Machine Learning, NLP, Embeddings, Vector Databases, RAG, Prompt Engineering

Frameworks & Libraries: Flask, PyTorch, TensorFlow, OpenCV, Ollama, REST APIs

Frontend: HTML, CSS, JavaScript, React.js

Databases & Platforms: SQLite, MySQL, Git, GitHub, VS Code, Jupyter Notebook

PROFESSIONAL EXPERIENCE

Software Development Intern

Vijayawada, AP

Vijesha IT Company Pvt. Ltd.

- Developed and debugged Python-based backend modules across multiple software projects.
- Improved application stability through structured testing and error handling.
- Participated in the complete software development lifecycle including requirements gathering, coding, testing, and deployment.
- Collaborated within an agile development team to deliver reliable software solutions.

PROJECTS

LANA – AI Emotional Support Assistant

2026

- Developed an AI-powered emotional support assistant using Python, Ollama, and Retrieval-Augmented Generation (RAG).
- Integrated real-time emotion detection using OpenCV and facial expression analysis.
- Built conversational memory supporting over 100+ multi-turn interactions.
- Implemented local LLM inference using Mistral through Ollama for privacy-focused AI conversations.
- Added speech synthesis and voice interaction capabilities for hands-free communication.
- Designed a scalable architecture combining Computer Vision, NLP, Vector Search, and Generative AI.

EDUCATION

Rise Krishna Sai Gandhi Group of Institutions (JNTUK)

Bachelor of Technology (Computer Science)

Expected Graduation: 2027

CGPA: 7.0

CERTIFICATIONS

- NPTEL – Artificial Intelligence Fundamentals (Silver Elite)
- Python 5-Star Problem Solver – HackerRank
- Software Development Training – Vijesha IT Company

ACHIEVEMENTS

- Built a complete AI assistant integrating LLMs, RAG, Voice AI, and Computer Vision.
- Earned NPTEL Silver Elite certification in Artificial Intelligence.
- Self-learned LLM architecture, embeddings, attention mechanisms, and vector databases.
- Developed production-style AI applications without relying on pre-built templates.